

Machine Learning Engineer – Job Description

Position

Machine Learning Engineer

Location

Pune, India (Hybrid/Remote)

Experience

1–4 Years

Employment Type

Full-Time

About the Role

We are looking for a passionate Machine Learning Engineer to design, develop, deploy, and optimize AI/ML solutions that solve real-world business problems. You will work closely with Data Scientists, Software Engineers, Product Managers, and stakeholders to build scalable machine learning systems and production-ready AI applications.

The ideal candidate has strong foundations in machine learning, deep learning, software engineering, and MLOps practices.

Key Responsibilities

Machine Learning Development

- Design, train, evaluate, and optimize machine learning models.
- Develop solutions for classification, regression, recommendation systems, forecasting, NLP, and computer vision tasks.
- Perform feature engineering, model selection, hyperparameter tuning, and performance optimization.
- Conduct experiments and analyze results to improve model accuracy and reliability.

Data Engineering & Processing

- Collect, clean, preprocess, and transform large datasets.
- Build data pipelines for training and inference workflows.
- Work with structured and unstructured data sources.

Model Deployment & MLOps

- Deploy machine learning models into production environments.
- Build scalable inference APIs and microservices.
- Implement monitoring, logging, model versioning, and retraining pipelines.
- Ensure model reliability, performance, and scalability.

Software Engineering

- Write clean, maintainable, and testable code.
- Collaborate with backend and frontend teams to integrate AI capabilities into applications.
- Participate in code reviews and architectural discussions.

Research & Innovation

- Stay updated with the latest developments in machine learning, deep learning, and generative AI.
- Evaluate new algorithms, frameworks, and tools.
- Prototype and validate emerging AI technologies.

Required Qualifications

Education

- Bachelor's or Master's degree in Computer Science, Artificial Intelligence, Data Science, Information Technology, or a related field.

Technical Skills

Programming

- Python (Strong proficiency)
- SQL
- Basic knowledge of JavaScript or Java

Machine Learning

- Supervised & Unsupervised Learning
- Ensemble Methods
- Model Evaluation Metrics
- Feature Engineering
- Time Series Analysis

Deep Learning

- Neural Networks
- CNNs
- RNNs/LSTMs
- Transformers
- Transfer Learning

AI Frameworks

- TensorFlow
- PyTorch
- Scikit-learn

Data Processing

- Pandas
- NumPy
- Apache Spark (Preferred)

MLOps

- Docker
- Git
- MLflow
- DVC
- CI/CD Pipelines

Cloud Platforms (Preferred)

- Amazon Web Services
- Google Cloud

- Microsoft Azure

Databases

- PostgreSQL
- MySQL
- MongoDB

Generative AI (Preferred)

- LLM Fine-Tuning
 - RAG Systems
 - Vector Databases
 - Prompt Engineering
 - Agentic AI Workflows
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Preferred Qualifications

- Experience deploying ML models to production.
 - Knowledge of distributed computing and scalable architectures.
 - Experience with LLMs and Generative AI applications.
 - Understanding of software engineering best practices.
 - Familiarity with Kubernetes and container orchestration.
 - Experience building end-to-end AI products.
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Sample Projects We Value

- Chatbots and AI Assistants
 - Retrieval-Augmented Generation (RAG) Systems
 - Recommendation Engines
 - Fraud Detection Systems
 - Computer Vision Applications
 - Forecasting and Predictive Analytics Platforms
 - MLOps Automation Pipelines
 - Autonomous AI Agents
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Soft Skills

- Strong problem-solving ability.
 - Analytical and critical thinking.
 - Excellent communication skills.
 - Ability to work independently and within cross-functional teams.
 - Curiosity and willingness to learn new technologies.
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Success Metrics

Within the first 6 months, you will:

- Deploy at least one production-grade ML model.
 - Improve model performance through experimentation and optimization.
 - Contribute to scalable ML infrastructure and deployment pipelines.
 - Collaborate effectively with engineering and product teams.
 - Deliver measurable business impact using AI solutions.
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Nice-to-Have Experience

- Fine-tuning open-source LLMs.
- Building AI agents using frameworks such as LangGraph and LangChain.
- Experience with vector databases such as Pinecone, Weaviate, or Qdrant.
- Working with multimodal AI systems.
- Experience building AI products serving thousands of users.

Expected Salary Range: ₹8–18 LPA (India, depending on experience and skill level).